

(a) Three meanings of proof geometry

Intrinsic

proof terms, paths, normalization

Girard; homotopy type theory



Structural

dependencies, modules, hyperedges

Mathlib network; universal proof graph



Learned

vectors, cosine neighborhoods, topics

formula/proof embeddings; retrieval

(b) What the three experiments add

I. Separate global views

tactic style vs premise domain

AMI 0.015



II. A local bridge

statement vs proof embeddings

AMI 0.080; rank r 0.341



III. An operational test

kernel-checked pass@3: 24% semantic / 21% BM25 / 16% none

separation → local coupling (Δ cosine 0.070) → intervention